Lauren Pothuru

+1 (603) 714-1351 | laurenpothuru100@gmail.com
LinkedIn | GitHub | Portfolio

SKILLS

Languages: Python, SQL, Java, C++, TypeScript
Tools/Libraries: PyTorch, Hugging Face, LangChain, AWS, PostgreSQL, NumPy/Pandas

EDUCATION

Cornell University, Ithaca, NY — Aug 2024– Dec 2027
Bachelor of Arts in Computer Science

- **GPA:** 3.6 / 4.0
- **Relevant Coursework:** Machine Learning, Algorithms, Data Structures, Linear Algebra, Discrete Math, Microeconomics, Operating Systems

EXPERIENCE

Software Engineer Intern, *Junior* – New York, NY — Incoming, Summer 2026

- Building LLM-powered tools used by investment teams at top global PE firms, MBB, Big 4, and investment banks.
- Owning a defined project from spec through deployment, with direct iterative feedback from senior consultants and investors.

AI Research Fellow, *Anote* – New York, NY — Incoming, Summer 2026

- Selected as 1 of 15 fellows for a 10-week in-person research program focused on evaluation frameworks, benchmarks, and agentic systems for frontier AI.
- Developing open-source evaluation benchmarks and contributing to a published research paper.

Software Engineer Intern, *InvestMint Inc.* – Toronto, ON, Canada — Dec 2025 – Present

- Architecting AWS-based ML infrastructure that ingests 10+ financial data sources to power cash-flow forecasting for small and mid-sized businesses.
- Reduced infrastructure costs by ~$1K/month through targeted vendor migration.

Teaching Assistant, *Cornell Bowers CIS* – Ithaca, NY — Aug 2025 – Present

- Mentor 260+ students in algorithm design, runtime analysis, and correctness proofs.
- Co-lead weekly discussion sections for 20+ students; design and grade problem sets emphasizing rigor and proof technique.

RESEARCH

Retrieval Perturbation as a Stress-Test for RAG Faithfulness, *Independent Research* — April 2026

- Designed a counterfactual benchmark to stress-test LLM behavior under controlled retrieval failure, applying two-proportion z-tests, Cohen's h, and Bonferroni correction across 2,400 evaluations on four open-weight models.
- Identified over-abstention (not hallucination) as the dominant failure mode and introduced a three-archetype failure taxonomy for evaluating production RAG systems; released as an open-source benchmark.

Bias Swap Audit: Counterfactual Auditing of Bias in Sentiment Models, *Independent Research* — Feb 2026

- Designed a counterfactual auditing pipeline that holds sentence content constant and swaps identity terms across six demographic categories, isolating model-level bias from contextual variation.
- Quantified statistically significant within-category gaps using Kruskal-Wallis, pairwise Wilcoxon, and Cohen's d; open-sourced as a toolkit that runs on arbitrary HuggingFace text-classification models.

PROJECTS

AI Credit Analysis Agent, *QuickFi* – Fairport, NY — Jan 2026 – May 2026

- Built a LangChain agent that validates credit applications against pulled credit records, generates risk assessments, and flags potentially fraudulent documents for a commercial equipment finance lender.
- Designed multi-stage pipeline covering document extraction, financial ratio calculation (DSCR, current ratio, debt-to-equity), and LLM-based reasoning over normalized financial data.